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CENTRE
NUMBER

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CANDIDATE
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CHEMISTRY

Grade 12

Paper 3

May 2025

1 hour 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Calculator
Qualitative Analysis Notes and Periodic Table
Ruler

12CHEM/03

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

You may receive fewer marks if you do not show your working or do not use appropriate units where required.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 30.

Answer all questions in English.

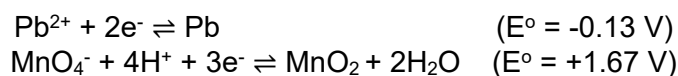
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This document consists of **7** printed and **1** blank page.

- 1 An experiment is set up to investigate how concentration affects the electrode potential of an electrochemical cell.

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The first experiment is performed under standard conditions. The half-equations of the electrochemical cell are shown.



- (a) Write the balanced ionic equation for the overall cell reaction.

.....

 [1]

- (b) (i) Draw a diagram of the apparatus you would use to measure the standard electrode potential for the lead/manganate(VII) ion cell. You should label all the electrodes and solutions.

[3]

- (ii) Calculate the standard cell potential (E_{cell}) of this cell.

$E_{\text{cell}} = \dots\dots\dots \text{V}$ [1]

(c) State the colour change during the experiment.

..... [1]

(d) The cell potential is measured for various concentrations of MnO_4^- (aq) and the results obtained are shown in the table below.

(i) Complete the third column of the table. Give each answer **two** decimal places.

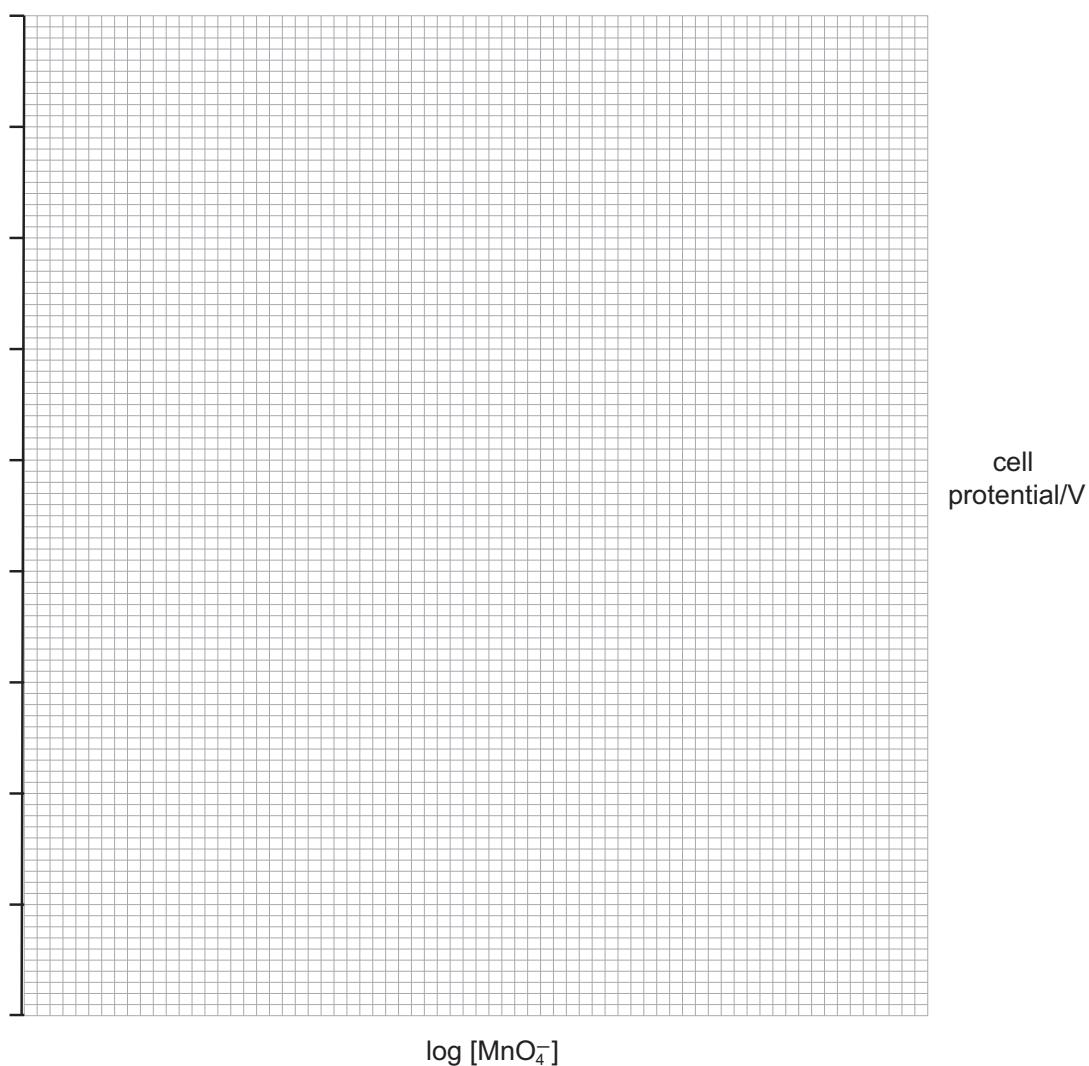
concentration of $\text{MnO}_4^-(\text{aq})/\text{mol dm}^{-3}$	cell potential / V	$\log[\text{MnO}_4^-(\text{aq})]$
5.00×10^{-1}	1.794	
1.00×10^{-1}	1.780	
4.00×10^{-2}	1.773	
1.00×10^{-2}	1.761	
5.00×10^{-3}	1.755	
2.00×10^{-3}	1.747	
8.00×10^{-4}	1.739	

[2]

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- (ii) Plot a graph to show the relationship between $\log [\text{MnO}_4^- (\text{aq})]$ and the cell potential measured **and** draw the line of best fit.

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[2]

- (iii) Use your graph to determine the concentration of $\text{MnO}_4^- (\text{aq})$ if the cell potential is 1.786 V.

concentration of $\text{MnO}_4^- (\text{aq})$ mol dm^{-3} [1]

- (e) State and explain how a decrease in concentration of Pb^{2+} affects the overall cell potential value.

.....

 [2]

- (f) The solutions contained in the two half-cells must be connected using a salt bridge.

Explain why potassium chloride solution is not suitable to use as the salt bridge.

.....
 [1]

- (g) A student conducts a second experiment using an electrochemical cell with the two half-equations shown.



The student observes no change in colour during the experiment.

Explain why $\text{Pb}^{2+}(\text{aq})$ will **not** reduce $\text{MnO}_4^{-}(\text{aq})$ to $\text{MnO}_2(\text{s})$.

.....

 [2]

[Total: 16]

- 2 A student is given four test-tubes containing colourless organic liquids labelled as **K**, **L**, **M** and **Q**.

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The liquids are:

- propanoic acid
- propanone
- propanal
- propan-2-ol.

The student uses experiments to identify the organic liquids.

- (a) In the first experiment pieces of magnesium ribbon are separately added to each test-tube, only **K** and **L** produce bubbles.
The second experiment is repeated with magnesium carbonate, only **K** produces bubbles.

- (i) Identify compound **K**

..... [1]

- (ii) Write a balanced equation of reaction of **K** with magnesium carbonate.

..... [1]

- (iii) Give positive test for gas produced with magnesium carbonate.

..... [1]

- (iv) Draw the structure of an isomer of **K** which does **not** release gas with carbonate ion.

[1]

(b) It is known that **L** and **M** can be oxidised.

(i) Suggest a suitable oxidising agent and give conditions.

Reagent [1]

Conditions [2]

(ii) Describe the observation of the reaction between **L** and oxidising in **b(i)**.

.....
..... [1]

(iii) Use the information from **(a)** and **(b)** to identify **L**.

compound **L** [1]

(c) Suggest the reagent which gives the same positive test with both **Q** and **M**.

..... [1]

(d) What would be observed when **Q** and **M** separately mixed with Tollen's reagent.

compound **Q**

compound **M**
[2]

(e) **Q** can be reduced to **L**.

(i) Give the formula of a suitable reducing agent.

..... [1]

(ii) Write a balanced equation for the conversion of **Q** to **L** using [H] to represent the reducing agent.

..... [1]

[Total: 14]

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